



IS222 – Database Management Systems

Semester I, 2025

Mode: Face to Face & Online (Only Online Submission Required)

Assignment 1 - Category: Individual Assignment

Due Date: [13th April, 2025]

Weight: 15%

Assignments Objectives

This assignment tests the following course learning outcomes and the associated CBOK. This assignment evaluates your ability to analyze business operations, design a relational database management system (RDBMS), and implement it using MS Access. The following learning outcomes will be assessed:

Attributes:

CLO 2: Construct a data model for a relational database management system.

1. Determine Identify and define business rules for a complex service-oriented enterprise.
2. Develop an Entity-Relationship Diagram (ERD) in Crow's Foot notation using MS Visio.
3. Create a comprehensive Data Dictionary, outlining data types, constraints, and relationships.
4. Implement the proposed database using MS Access, ensuring all relationships, constraints, and dependencies are correctly applied.
5. Build SQL queries, forms, and reports to analyze and retrieve insights from the dataset.

***Note:** The excel marking rubric will be used to assess this assignment and is uploaded onto Moodle.

Prerequisites

Each student is expected to complete this assignment on their own; that is, individually and NOT in groups.

Before you commence this assignment, you should have at least completed:

- Reading chapters 1-4 of the textbook.
- Completed all required Moodle (Course) activities from Weeks 1 to 5.
- You should have covered all materials and labs exercises on designing databases with MS- Access and MS Visio 2016. Assignment Overview

A Case Study:

SuvaTech Solutions Database System

SuvaTech Solutions, a leading Fijian IT consultancy firm based in Suva, provides specialized services across multiple sectors: Government, Enterprise, and SMEs. The company works closely with local businesses, ministries, and international organizations, offering IT infrastructure management, software development, and AI-driven solutions. Each client organization signs contracts that include specific service-level agreements (SLAs), project scopes, renewal terms, and compliance requirements. Clients can have multiple contracts, but each contract must be linked to a specific business division within the company.

Each consultant is assigned to projects based on their expertise level (Junior, Senior, or Principal) and security clearance (Confidential to Top Secret). Consultants may work on multiple projects but must meet the required certifications and security clearance levels before being assigned. Consultants must undergo training sessions to maintain their certifications, and expired certifications prevent participation in specific projects.

The organization also manages cloud service and software licenses; which clients subscribe to under different pricing tiers. Clients can upgrade, downgrade, or renew their subscriptions, which impacts billing cycles, user limits, and service access.

Requirements: Part A Database Design Report in PDF Format

Ensure to follow the Assignment 1 requirements prudently.

Part A - Database Design Report in PDF Format

Submit Part A as a PDF with the answers for each part below under separate sections/headings.

The sections are as follows:

1. Cover Page

- INCLUDE the ID number and full name, Lab Day and Time, and Campus.

2. Table of Contents

3. Introduction

Introduce the business process, the role of the database, and the advantage(s) of using the database for this given scenario above.

4. Business Rules (BR)

- From the scenario given, identify the business rules (BR). The BRs should lead directly to your ERD and capture concisely the entities/relationships in the rule and should reference any underlying assumption for its basis (see third bullet point)
- Briefly explain each business rule, that is, explain the importance of each business rule to your database. You should have a valid reason for including the business rule.
- If you are making any assumptions as a basis for any of your business rules, create a separate section called “Assumptions” under this heading, and list your assumptions clearly.

5. Entity Relationship Diagram (ERD)

- Create an Entity Relationship Diagram (ERD) using Crow’s Foot notation in MS Visio. Label the relationships using appropriate verbs. Insert this diagram into your report.

6. Data Dictionary

- Create a data dictionary for your database. A sample Data Dictionary is on Moodle beneath the Assignment 1 specifications link. Follow the same format as in the sample.

7. Conclusion

- Discuss what you have achieved in this assignment, the resources that you have used to complete this assignment, and the difficulties that you have faced while doing the assignment.

Supplementary checklist to rest assured in conjunction with the expectations for Part A

- Content Coherence: The report content is clear, accurate, and relevant to all requirements. All placements are logically accurate.
- Organization: (including submission time) Arrangements are well assured, appropriately sectioned, and important points are emphasized appropriately.
- Creativity in the portrayal of information obtainable.

Part B - MS Access Database (DB)

This section focuses on building your consultancy management database in MS Access. You will work through the steps of creating tables, inserting data, building queries, generating reports, and developing data entry forms all based on the scenario involving clients, consultants, projects, contracts, certifications, and cloud services.

1. Create Tables with correct attributes, primary keys, and foreign keys etc.
2. Populate Tables with at least 30 records for consultants, 50 for contracts, and 100 for projects. [Failure in following the instructions will result in a penalty.]

Hint: Create the necessary tables in MS Access using your ERD (Entity Relationship Diagram) and Data Dictionary.

Each table must include:

- Correct attributes (fields) with appropriate data types.
- Defined primary keys.
- Properly linked foreign keys for relationships.

3. Create queries to respond to the following inquiries: Hint: Use filters, sorting, totals, and criteria options as needed.

Answer the following business questions using Access queries:

- Q1. Retrieve all contracts linked to a specific client.
- Q2. Identify consultants who have expired certifications.
- Q3. Find the most requested specialization used in consultancy projects.
- Q4. List all projects that do not meet security clearance requirements.
- Q5. Identify clients who have upgraded their cloud service subscriptions in the last quarter.

4. Create Reports in MS Access

- Use the Report Wizard or Report Design to create: A summary report showing the number of active contracts per sector (e.g., Healthcare, Finance, Education).
- A consultant-wise certification report that helps track upcoming or expired certification renewals.

5. Develop user-friendly forms for data entry:

- **Client Registration Form**
 - To add new client details such as name, email, contact number, and organization.
- **Project Assignment Form**
 - To assign a consultant to a project, including fields like project name, consultant name, assignment date, and specialization.

Submission Guidelines:

- Submit on Moodle, upload the Database Design Report (part A) PDF file, and the MS Access file (part B).

For Example:

SXXXXXXX AI Report.pdf (if it's a pdf file),

SXXXXXXX AI DB.accdb (if it's your student ID) is shown by the Xs.

LATE SUBMISSION = ZERO (0) **Plagiarism**

- Incorrect submissions will result in loss of marks or simply a grade of zero being awarded.
- Late Submission = Grade of Zero (0) for Assignment 1.

Zero for any penalty cases thus no two assignment submissions should produce the same results. Any such case will be dealt with severely, with actions including awarding a mark of zero, forwarding the case to the Head of School, and reporting the matter to the Student Disciplinary Committee.

Queries

- Maintain contact through Moodle discussion forum named Assignment 1 Discussion Forum
- Feel free to make an appointment via email with the Course Coordinator for any urgent matters regarding Assignment 1.

The End...

Assignment 1 Assessment Rubric					
IS222 – Database Management Systems – S12025					
Assignment 1 Marking Rubric					
Marker:		Student ID:		Total Mark:	0.00
Part A – Database Design Report in PDF format (60 marks) – CBOK (A, B, C)					
Question	Details	Unsatisfactory	Satisfactory	Good	Marks
Introduction – (5 marks)	Summary of the case study (2)	0 – 0.5	1 – 1.5	2	
	Role of the database (1)	0	0.5	1	
	Advantage(s) of using the database (2)	0 – 0.5	1 – 1.5	2	
Business Rules – (20)	Business rules capture the business problem	0 – 7	7.5 – 10.5	11 – 15	
	Rules are clear and well-formatted (5)	0 – 2	2.5 – 3.5	4 – 5	
ERD – MS Visio (20 marks)	ERD reflects business rules (4)	0 – 1.5	2 – 3	3.5 – 4	
	Appropriate entities (3)	0 – 1	1.5 – 2.5	3	
	Relationships & types (2)	0 – 0.5	1 – 1.5	2	
	Entity names (2)	0 – 0.5	1 – 1.5	2	
	Correct relationship types (2)	0 – 0.5	1 – 1.5	2	
	PKs and FKs identified (2)	0 – 0.5	1 – 1.5	2	
	Attributes listed (2)	0 – 0.5	1 – 1.5	2	
	Diagram rules followed (2)	0 – 0.5	1 – 1.5	2	
	Identifying/non-identifying (1)	0	0.5	1	
	Format & metadata (1)	0	0.5	1	
Data Dictionary (5 marks)	Correct data types (2)	0 – 0.5	1 – 1.5	2	
	PK/FK stated (1)	0	0.5	1	
	NULL/NOT NULL fields (1)	0	0.5	1	
	Achievements (2)	0 – 0.5	1 – 1.5	2	
Conclusion (5 marks)	Resources used (1)	0	0.5	1	
	Difficulties faced (2)	0 – 0.5	1 – 1.5	2	
	Sectioning & headings (2)	0 – 0.5	1 – 1.5	2	
Presentation (5 marks)	Format consistency (2)	0	0.5	1	
	Page numbers/layout (1)	0	0.5	1	
	Figure labels (1)	0	0.5	1	
Part A Total					0
Part B – MS Access Database (60 marks) – CBOK (A)					
Question	Details	Unsatisfactory	Satisfactory	Good	Marks
Tables & Attributes (15)	Entities created (5)	0 – 2	2.5 – 3.5	4 – 5	
	Attributes & Data Types (5)	0 – 2	2.5 – 3.5	4 – 5	
	Matches Data Dictionary (5)	0 – 2	2.5 – 3.5	4 – 5	
Relationships (5 marks)	Relationship diagram (5)	0 – 2	2.5 – 3.5	4 – 5	
Forms (5 marks)	Data entry forms usable (5)	0 – 2	2.5 – 3.5	4 – 5	
Data Entry (5 marks)	Realistic sample data (5)	0 – 2	2.5 – 3.5	4 – 5	
Queries (24 marks)	Q1-Q5 logic and grouping	0 – 8	9 – 16	17 – 24	
Reports (10 marks)	Active contracts per sector (5)	0 – 2	2.5 – 3.5	4 – 5	
	Consultant certification report (5)	0 – 2	2.5 – 3.5	4 – 5	
File Submission (5 marks)	All files submitted & open (5)	0	2	3.5 – 5	
Part B Total					
Overall Total					0